The Art of Uncertain Topoplots: Designing and Evaluating Uncertainty Visualizations for EEG/ERP Data

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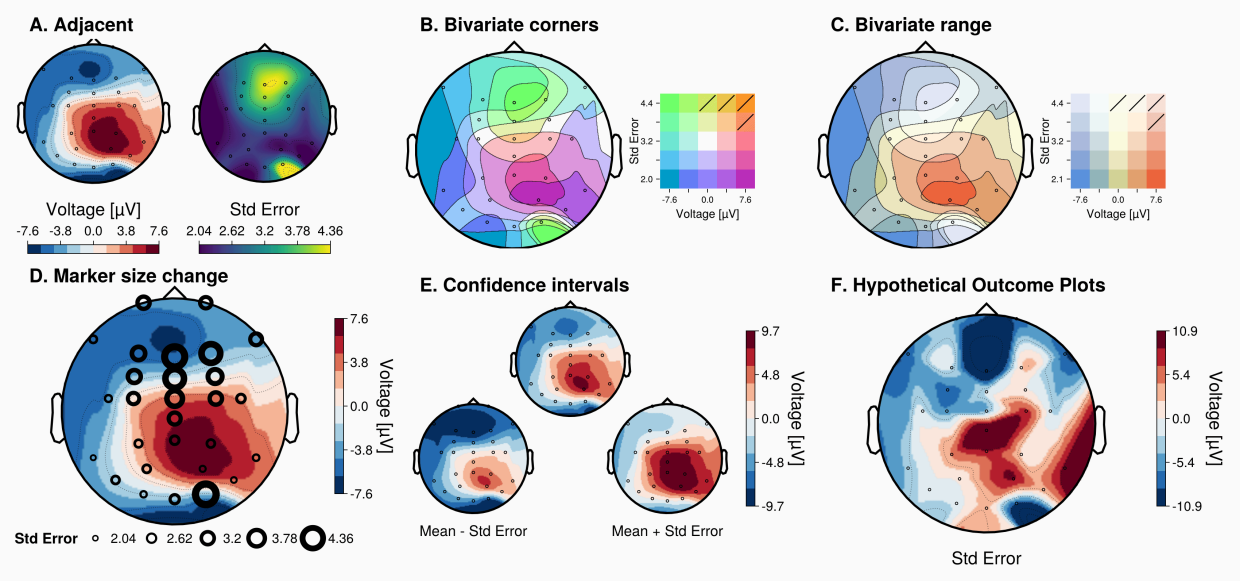


Figure 1: Overview of the evaluated uncertainty-visualization designs.

Abstract—TODO: Abstract

Index terms—Data visualization, Electroencephalography, Human factors, Uncertainty

1 Introduction

Scientific estimates are often accompanied by uncertainty, but representing uncertainty for spatial and multivariate data is challenging task. For individual estimates and curves, familiar techniques such as error bars and confidence bands provide relatively direct ways to communicate uncertainty [32]. But in spatial visualizations the estimated signal and its uncertainty vary across space and need to be shown at the same time.

EEG/ERP topographic plots are a useful case for studying this problem. EEG measures electrical potentials at electrodes distributed across the scalp. ERP or event-related potentials are averaged and time-locked EEG trials. A topographic plot or **topoplot** is a figure with ERP over a two-dimensional representation of the head, often interpolated. Colour usually represents the sign and magnitude of the estimated voltage, allowing researchers to inspect the spatial distribution of signals. Topoplots are also one of the most familiar visualization types in the EEG/ERP community. In our previous survey of 213 M/EEG researchers, 52% of respondents reported having previously created topoplots. This is the second results after conventional ERP waveforms (61%) [22].

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Manuscript received DD MMM. YYYY; accepted DD MMM. YYYY. Date of Publication DD MMM. YYYY; date of current version DD MMM. YYYY. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.YYYY.xxxxxxx

Conventional topoplots generally show the estimated voltage without an accompanying measure of its precision. There is no established visual convention for integrating uncertainty into these maps. Although there are some software solutions for representation of uncertaint in EEG visualization [9].

Showing uncertainty could help researchers distinguish regions with relatively stable estimates from regions with larger variability and provide a visual impression of the local signal-to-noise structure. On the other hand, adding another visual variable to an already familiar spatial representation may hinder the perception of the underlying ERP pattern. This creates a broader visualization question: **how can uncertainty be added to an established spatial visualization while preserving the primary signal that users already know how to read?**. This difficulty is reflected by [1]: they reviewed 1,451 figures from six neuroscience journals, finding that uncertainty was particularly underrepresented in visualizations of data with more than three features, with only 20% indicating uncertainty.

Before the quantitative evaluation, we conducted qualitative interviews with nine EEG/ERP experts to understand how topoplots are used in practice, how researchers think about uncertainty, and what they expect from an uncertainty-aware representation. The interviewed researchers valued uncertainty representation on topoplots but reported a lack of clear conventions or guidance for doing so. These findings helped us formulate the research gap, select the visualization approaches, and design the later experimental tasks.

Building on prior uncertainty-visualization work and our qualitative interviews with EEG/ERP experts, we created several uncertainty representations and compared them with the conventional topoplot in a user study. We tested whether showing uncertainty changes participants' ability to recognize the underlying spatial ERP pattern, how well different representations support judgments that require combining

Field	Study	Data sources	Main finding
Neuroscience	Allen et al. [1]	288 articles; 1,451 figures	Only about 20% of visualisations with three features represented uncertainty.[1]
Ecosystem science	Yanai et al. [33]	132 articles	Studies reported numerical estimates for only 42% of relevant uncertainty sources on average; 54% of studies reported such estimates for fewer than half.
Geospatial / public-health mapping	Jain et al. [14]	66 articles containing maps of climate-related infectious diseases in the Asia-Pacific region	Only 3 of 66 studies (4.5%) visually integrated uncertainty into their maps; 26 studies (39%) did not report uncertainty at all.
Visualization software	Petek et al. [25]	5 visualization tools (Matplotlib, ggplot2, ggdist, Seaborn, and ggdibbler); 18 example uncertainty-visualization scenarios	In 13 of 18 scenarios (72%), all five tools required users to quantify uncertainty and construct its visual encoding manually.

Table 1: Representative evidence of incomplete uncertainty reporting and visualization across scientific domains and limited software support.

signal and uncertainty, and how participants experience and interpret the different designs. The study combines formative expert interviews, controlled performance evaluation, subjective ratings, and open-ended qualitative explanations of participants' judgments and decisions.

To our knowledge, this is the first systematic study to compare multiple uncertainty-visualization strategies for EEG/ERP topoplots in a controlled user study. While uncertainty has occasionally been represented in EEG visualization, there is no established practice for integrating it into scalp topographies, and the relative effectiveness of alternative representations has not been systematically evaluated.

Our contributions are fourfold. First, we report qualitative insights from EEG/ERP experts on current topoplot practices and the challenges of communicating uncertainty in them. Second, we created a design pool of uncertainty representations for EEG/ERP topoplots. Third, we provide empirical evidence of how different plot types affect participants' ability to interpret the underlying spatial ERP pattern and make judgments that combine signal and uncertainty. This is complemented by participants' subjective ratings and qualitative feedback explaining their preferences and interpretation difficulties. Finally, we derive practical recommendations for the usage of uncertainty-aware EEG/ERP topoplots.

2 Related Work

2.1 Uncertainty Visualization in Scientific and Spatial Data

Research on visualizing uncertainty in spatial data extends back several decades. An early contribution is MacEachren's 1992 discussion of cartographic symbols and interfaces for communicating uncertainty during exploratory analysis [21]. By 2009, Sanyal et al. had published a controlled comparison of uncertainty encodings for one- and two-dimensional datasets [26]. In 2012, Brodlie et al. reviewed uncertainty visualization across a broader range of scientific applications [4]. These works illustrate both the development of representations and efforts to evaluate their interpretation.

Subsequent contributions refined the characterization and communication of uncertainty. Hullman et al. introduced hypothetical outcome plots in 2015 [13], and Correll et al. presented Value-Suppressing Uncertainty Palettes in 2018 [5]. More recent work continues to investigate these questions in specific scientific domains, including Sterzik et al.'s 2025 evaluation of biomolecular uncertainty visualizations [28]. These examples show sustained interest in both representation design and empirical evaluation across scientific domains.

Across this literature, representations differ in whether viewers encounter numerical uncertainty summaries, intervals, or possible outcomes [32]. The following subsections examine reporting practices, encoding choices, and the evidence about their use before considering the remaining gap in EEG/ERP visualization.

2.2 Uncertainty Is Often Missing from Scientific Reporting and Visualization

Uncertainty is routinely omitted or incompletely communicated across scientific domains. Boukhelifa and Duke describe a persistent divide between the recognized importance of uncertainty in visualization research and its limited adoption in practice [3]. Similarly, Hullman's survey of visualization authors found that, although authors generally acknowledged the value of communicating uncertainty, direct uncertainty depiction was routinely omitted [12]. Table 1 summarizes this observation with evidence from neuroscience, ecosystem science, geospatial mapping, and visualization software.

Limited software support is one practical reason why uncertainty visualizations may remain uncommon. Although many uncertainty visualization techniques have been proposed, they are not necessarily available as standard functionality in commonly used visualization systems. Petek et al. compared five visualization tools across 18 example uncertainty visualizations and found that, for 13 of these examples, all five tools required users to quantify uncertainty themselves and implement its visual encoding through custom plotting code, rather than relying on dedicated uncertainty-visualization functionality (their Table 3) [25].

This software gap has been reported across multiple visualization ecosystems and application domains. These observations were not based on reported systematic audits of software packages, but together they indicate a recurring gap between uncertainty-visualization research and routinely available software functionality. Kay notes that support for uncertainty visualization in existing Grammar-of-Graphics implementations remains largely limited to conventional representations such as error bars, uncertainty bands, and density plots, while many techniques proposed in visualization research remain difficult to construct [16]. In geospatial visualization, Koo et al. identify a lack of uncertainty-visualization tools in standard GIS environments and implement dedicated uncertainty encodings as an ArcGIS extension [18]. Earlier work in GIS likewise observed that, despite a broad conceptual literature on uncertainty visualization, implemented tools and pilot projects remained rare [19]. In medical visualization, Gillmann et al. further reported that, to their knowledge, no commercially available uncertainty-aware visualization technique was available in medical workstations [11].

2.3 Visual Encodings for Spatial Uncertainty

Kinkeldey et al. [17] classify spatial uncertainty visualizations along three dimensions: **intrinsic versus extrinsic** (modifying existing visual properties or adding graphical elements), **coincident versus adjacent** (showing data and uncertainty together or separately), and **static versus dynamic** representations. We focus on five approaches used in our study.

[1] In Allen et al.'s terminology, three-dimensional graphics represent three data variables, which can be shown in a flat figure using two axes and colour; the term does not necessarily imply a 3D spatial rendering [1].

Markers and glyphs represent uncertainty through graphical elements added to a spatial display. Their visual properties, such as size, colour, or opacity, can encode uncertainty independently of the underlying data [4, 21].

Bivariate colour maps encode two quantities within a single spatial view using a two-dimensional colour legend [29, 31]. MacEachren [21] proposed using this approach to display data values together with their uncertainty, assigning different colour properties to each quantity.

Adjacent or juxtaposed maps display an estimate and its uncertainty in separate, spatially aligned views [4]. Each quantity can use its own colour scale, allowing viewers to compare the two maps.

Interval-based representations display an estimate together with its lower and upper bounds. For spatial data, these bounds can be shown as separate maps alongside the estimate [9]. Depending on the statistical method, the bounds may represent confidence intervals or another interval, such as the mean plus or minus one standard error.

Hypothetical outcome plots use animation to display samples from a modelled distribution [13]. For spatial data, each frame shows a possible spatial pattern, communicating uncertainty through variation across successive frames.

2.4 Evaluating Uncertainty Visualizations

Uncertainty visualizations have been evaluated using different perceptual and decision-making tasks. MacEachren et al. [21] compared adjacent maps, texture overlays, and bivariate colour encodings using tasks involving spatial patterns and data reliability. Sanyal et al. [26] compared error bars, glyph size, glyph colour, and surface colour using search and counting tasks on one- and two-dimensional datasets. Both studies demonstrated that the relative effectiveness of uncertainty representations depends on what participants are asked to identify or compare.

Other studies have examined whether viewers can correctly interpret and use uncertainty. Hullman et al. [13] compared hypothetical outcome plots with error bars and violin plots using probability judgments about the ordering of variables. Correll et al. [5] evaluated bivariate colour maps and Value-Suppressing Uncertainty Palettes using both identification tasks, which required participants to decode values and uncertainty, and decision-making tasks, which required them to consider both quantities. More recently, Sterzik et al. [28] evaluated biomolecular uncertainty visualizations through expert assessments and perceptual experiments, examining intuitive interpretation, perceived mapping direction, and accuracy in estimating uncertainty.

Evaluation measures capture different aspects of visualization performance. Accuracy indicates whether participants reach the correct interpretation, while response time measures how quickly they complete a task. Subjective ratings assess perceived difficulty, usefulness, or preference, and qualitative feedback can help explain why participants favour or struggle with particular designs [20].

Results from different evaluation methods do not always agree. Reviewing previous user studies, Kinkeldey et al. [17] found that visualization effectiveness depends on the task and that subjective preferences do not necessarily correspond to better objective performance. This motivates our use of two performance tasks alongside subjective ratings and qualitative feedback to evaluate uncertainty visualization techniques for EEG topographic maps.

2.5 Uncertainty in EEG/ERP Visualization

Uncertainty is commonly represented for one-dimensional ERP waveforms using standard errors or confidence intervals, displayed as error bars or shaded bands [1, 15, 24]. Statistical scalp maps and significance markers are also familiar in EEG analysis. These indicate evidence from hypothesis tests, such as test statistics or thresholded significance, and do not directly show uncertainty in the estimated voltage itself [24].

LIMO EEG provides bootstrap confidence intervals for estimates at individual electrodes and time points [24]. Its published examples typically separate statistical topographies from interval displays: scalp maps show test statistics, while waveforms for selected electrodes show estimated responses or model coefficients with confidence intervals (their Fig. 4). This supports inspecting statistical evidence across the scalp and uncertainty in selected time courses.

The R package `diegr` provides a particularly relevant precedent [9]. Its `plot_topo_mean()` function displays three scalp maps corresponding to the lower confidence bound, mean, and upper confidence bound. It therefore provides a direct example of interval-based uncertainty visualization in EEG topographies. Neither the LIMO EEG paper nor the `diegr` documentation reports a systematic perceptual or user evaluation comparing alternative uncertainty encodings for EEG/ERP topoplots.

These individual implementations demonstrate prior use of uncertainty displays without establishing a shared convention for EEG/ERP topoplots. Their relevance here concerns uncertainty in estimated sensor-space voltage. Interpolation accuracy and confidence regions for source-localization estimates concern different targets of uncertainty and do not by themselves provide an encoding of voltage uncertainty alongside the scalp signal.

2.6 Research Gap

The literature provides established uncertainty-visualization techniques and individual EEG implementations, but little comparative evidence for EEG/ERP topoplots. To our knowledge, the alternative strategies have not been systematically compared in a controlled user study in this context, and no established visual convention guides their use. The open question is which representations preserve the underlying spatial signal while supporting uncertainty-aware judgments. Addressing this question requires evidence about performance as well as participants' explanations of why they prefer particular representations or struggle to interpret them.

3 Formative Qualitative Study

Before adapting and quantitatively evaluating the uncertainty visualizations, we conducted a formative qualitative study to understand how EEG/ERP researchers currently use topoplots, how they conceptualize uncertainty, and which barriers they encounter when communicating it in spatial EEG representations. The interviews also served to identify relevant user tasks and requirements for the subsequent visualization design and quantitative study.

3.1 Participants

We conducted semi-structured interviews with nine participants with experience in EEG/ERP research or clinical EEG. The sample covered different application contexts, including fundamental neuroscience, brain-computer interface research, and clinical EEG practice.

3.2 Interview Guide

We used a semi-structured interview guide consisting of 14 main questions. A shortened version containing the main questions was provided to participants, while the interviewer used an extended version containing optional probes.

The interview covered six main topics: participants' EEG/ERP background and visualization practices; their understanding and current use of uncertainty; the purposes and audiences for which they use topoplots; current and potential approaches to showing uncertainty on topoplots; perceived value and barriers to uncertainty visualization; and requirements for future topoplot tools.

Questions were deliberately open-ended, and the order and depth of follow-up questions could vary depending on the participant's responses. The interviewer maintained a neutral stance and did not suggest that uncertainty should or should not be visualized.

3.3 Procedure and Transcription

The interviews lasted approximately 15–30 minutes and were audio-recorded. At the beginning of each interview, participants explicitly consented to the recording, to public sharing of an anonymized transcript for research purposes, and to the use of anonymized quotations in publications.

Audio recordings were transcribed using Voice AI, the speech-to-text service provided through the Academic Cloud/GWDG [10]. Transcripts were anonymized before analysis and publication.

3.4 Analysis and Implications for the Visualization Design

The interviews were analyzed using a structured qualitative content-analysis approach. Responses were first organized according to the interview questions and then compared across participants to identify recurring practices, concerns, and requirements.

Participants saw value in communicating uncertainty, but rarely visualized spread-related uncertainty directly on topoplots. They reported ambiguity about how to represent it, limited support in the software they used, and a lack of clear conventions or guidance. Together with the individual implementations described in the related work, these findings motivate evaluating alternative representations and developing practical guidance for communicating uncertainty spatially.

Participants' accounts referred both to variability or precision, such as standard deviation or standard error, and to statistical evidence or signal reliability, such as signal-to-noise ratios, test statistics, or significance values. These quantities answer different questions: significance does not directly describe the precision of a voltage estimate. We focused the subsequent study on representing standard error alongside estimated voltage.

Several recurring uses of topoplots were identified, including recognizing expected spatial ERP patterns, comparing experimental conditions, assessing plausible spatial distributions, and locating relevant signals. We used these common practices as the basis for constructing the tasks in the quantitative study.

Participants also proposed possible uncertainty representations. For example, R2 suggested showing three topoplots representing an uncertainty interval together with the estimate. This suggestion helped motivate the inclusion of an interval-based representation with precedent in EEG visualization. We combined these interview-derived requirements with existing visualization approaches to construct the design pool evaluated in the subsequent study.

4 Design Space of Uncertainty Visualizations

We selected six approaches for representing standard error alongside estimated voltage by combining designs from prior visualization work with ideas from the formative interviews. The two bivariate topoplot designs were inspired by the range and corners models described by Strode et al. and by Correll et al.'s Value-Suppressing Uncertainty Palettes [5, 29]. The resulting bivariate topoplots are our original designs: we used these color-model concepts as starting points and adapted them to spatially continuous EEG/ERP topoplots that jointly encode signed voltage and standard error. Adjacent plots, uncertainty markers, and hypothetical outcome animation were also informed by existing visualization approaches and adapted for our study. The interval-based representation was suggested by participant R2 and has precedent in the three-map display provided by `diegr` [9]. We included a conventional topoplot without uncertainty information as a baseline.

The designs differ in whether uncertainty is presented in separate maps, integrated into color, overlaid as markers, or conveyed through animation. This selection allowed us to examine how these approaches support two needs identified in the interviews: preserving recognizable spatial patterns and supporting judgments that combine the estimated signal and its uncertainty.

We aimed to use RdBu or similar diverging red–blue color schemes for voltage wherever possible, to maintain visual consistency across designs. We used discrete (binned) colormaps, motivated by Padilla et al.'s study of grayscale 2D scalar fields [23], which found improved accuracy for value lookup and highest-point identification under some binning conditions, while continuous encodings were often faster. For univariate voltage maps, we quantized RdBu into 10 colors. Both bivariate designs used a 5×5 grid, with five bins each for voltage and standard error, yielding 25 color combinations.

4.1 Basic topoplot

The basic topoplot served as the baseline condition. It displays only the estimated EEG/ERP voltage amplitude using a diverging color scale, such that color encodes both the sign and magnitude of the signal. No information about standard error (SE) is shown.

4.2 Adjacent plots

The adjacent-plots design was inspired by the EEGVIS toolbox [7]. It presents the estimated signal and its SE in two separate topoplots. The signal map uses a diverging color scale to represent negative and positive voltage amplitudes, whereas the SE map uses a sequential color scale because SE is non-negative. This design preserves a familiar univariate encoding for each quantity, but requires viewers to compare corresponding spatial regions across the two maps.

4.3 Bivariate color: corners

The corner encoding adapts the corners model described by Strode et al. [29], using the TeulingFig3 four-corner, two-dimensional color legend [30] to jointly represent signal and SE.

The corners model is intended for analyses that emphasize extreme combinations of two variables while de-emphasizing intermediate values [29]. This makes it suitable for our study tasks, in which viewers must locate and compare regions combining strongly positive or negative voltage with low or high uncertainty.

We selected this palette because it received the highest assessment category for elementary localization and identification (ER1), equal visual importance (ER2), and elementary comparison (ER3) on dark backgrounds in Bernard et al.'s task-based evaluation [2]. Its corners use distinct hues so that the four joint extrema can be identified and compared. We further adjusted the palette to improve separation between neighboring combinations, following the broader warning that blue–violet mixtures in bivariate maps can be difficult to distinguish [29]. Hatched legend cells mark combinations absent from the displayed topoplot.

4.4 Bivariate color: range

The range encoding adapts the range model described by Strode et al. [29] while retaining the familiar cool-to-warm diverging organization of conventional EEG/ERP topoplots.

The range model is intended for analyses in which viewers examine the range of one variable within levels or categories of another [29]. In our application, it supports assessing how SE varies within regions of positive or negative voltage while preserving the spatial signal pattern needed for comparison.

Voltage is represented along a diverging color dimension, with cool and warm hues indicating opposite polarities. SE is encoded along the second dimension by progressively increasing lightness and decreasing chroma while approximately preserving the hue associated with each voltage level. Consequently, regions with greater uncertainty appear lighter and less saturated, while the underlying color identity of the voltage remains recognizable. As in the corners design, hatched legend cells indicate color combinations absent from the displayed topoplot.

4.5 Interval-based representation

The interval-based condition uses three adjacent maps showing the estimated mean, the mean minus one SE, and the mean plus one SE. In our implementation, the bounds indicate $\pm$ one SE rather than a 95% confidence interval. Viewers compare the spatial voltage distribution across the three maps: larger differences indicate greater SE, while smaller differences indicate more precise estimates.

This visualization has precedents in both EEG and fMRI. For EEG, the R package `diegr` provides a `plot_topo_mean()` function that uses adjacent topographic maps of the mean and its lower and upper confidence bounds [9]. For fMRI, Engel and Burton proposed an analogous three-map representation showing an estimated activation map together with its lower and upper confidence bounds [8].

4.6 Uncertainty markers

The uncertainty-marker design overlays spatial markers on the voltage topoplot and maps SE to marker size. Larger markers indicate higher SE and smaller markers indicate lower SE. This separates the visual channels used for the two quantities: color represents the estimated signal, while size represents uncertainty.

Marker size is an established visual channel for quantitative uncertainty [4, 21]. We included this condition to examine how an added size encoding affects reading of the underlying voltage pattern and its uncertainty.

4.7 Hypothetical outcome animation

This condition adapts the hypothetical-outcome approach of Hullman et al. to spatial EEG/ERP data [13]. We generated a finite sequence of plausible topoplots from the estimated signal and its SE and displayed them as an animation with a stable visual mapping across frames. Each frame depicts one hypothetical outcome, not a successive point in time. Regions that remain visually similar across frames are comparatively stable, whereas regions that vary strongly indicate greater uncertainty.

5 Quantitative User Study

5.1 Research Questions

The quantitative study addressed three research questions:

RQ1. Pattern comparison: To what extent do uncertainty visualizations preserve the ability to recognize the underlying spatial pattern in an EEG/ERP topoplot?

RQ2. Uncertainty integration: How well can users integrate the estimated signal and its uncertainty to identify spatial regions with a high signal-to-noise ratio, and how does performance depend on the visualization technique?

RQ3. Subjective evaluation: How do users subjectively evaluate the different approaches to visualizing uncertainty in EEG/ERP topoplots?

5.2 Participants

We recruited participants with experience in EEG/ERP research or analysis. The study targeted researchers and students familiar with EEG/ERP data and topographic visualizations. Participation was voluntary.

A total of [N] participants were included in the quantitative analysis. *TODO: Add demographics, EEG experience, and exclusion criteria.*

5.3 Stimuli

We used three characteristic ERP topographic patterns: mismatch negativity (MMN), P3, and N170 [15]. For each pattern, we prepared a pool of ten pattern variations, from which individual stimuli were selected during the experiment.

The patterns were displayed using seven visualization conditions: a basic topoplot and six uncertainty visualizations selected from the design pool described above. Each visualization technique was combined with each of the three ERP patterns, forming a 7×3 within-participant design.

All visualization types were implemented in Julia using Makie [6]. We used this implementation to render the static plots and hypothetical-outcome animations presented as stimuli in the online survey.

5.4 Tasks

5.4.1 Task 1: Pattern comparison

Task 1 tested whether adding uncertainty information interfered with recognition of the underlying spatial pattern. In each trial, participants saw a target ERP pattern and two alternative topoplots and selected the alternative that matched the target most closely. One alternative belonged to the same ERP-pattern class as the target, whereas the other was drawn from one of the two remaining classes.

Each visualization technique was shown once with MMN, P3, and N170, resulting in 21 trials. Each trial was shown on a separate page, and trial order was randomized for each participant. For each trial, the correct and distractor pattern variations were sampled from their respective stimulus pools. The distractor class was randomly selected from the two non-target classes. The position of the correct response alternated between the first and second alternative across the predefined trials.

5.4.2 Task 2: Uncertainty integration

Task 2 tested whether participants could combine information about the estimated voltage and its uncertainty to identify a relevant spatial region. On each trial, participants saw one topographic visualization and were asked to select the region with the highest signal-to-noise ratio (SNR), defined in the study as

$$\mathrm{SNR} = \left| \frac{\text{Voltage}}{\text{Standard Error}} \right|. \tag{1}$$

Responses were collected using a separate 64-region selectable scalp layout displayed next to the stimulus. We constructed this layout as a fixed SVG representation of the scalp divided into 64 individually selectable polygonal regions, labelled internally from R01 to R64. The polygon layer was aligned with a scalp-montage reference, and the same spatial layout was used for every trial. Participants selected one region by clicking directly on the corresponding polygon.

As in Task 1, the complete design contained each combination of the seven visualization techniques and three ERP patterns, giving 21 trials, each presented on a separate page. For each ERP pattern, seven distinct variations were randomly selected from its pool and assigned across the seven visualization techniques. Trial order was randomized for each participant.

5.4.3 Task 3: Subjective evaluation

After completing the objective tasks, participants evaluated each of the seven visualization techniques subjectively. Four aspects were assessed on 5-point Likert scales: how helpful the visualization was for comparing spatial patterns, how helpful it was for detecting relevant spatial regions, how helpful it was for representing signal and uncertainty, and how visually appealing it was.

5.5 Procedure

The study was conducted online. Before beginning the study, participants were informed about data anonymisation, public data sharing, voluntary participation, and their right to withdraw, and provided informed consent. Participants then answered background questions concerning their experience with EEG/ERP analysis.

Participants then received an explanation of the visualization techniques and completed Task 1 (pattern comparison), followed by brief feedback questions. They next received instructions for Task 2 (uncertainty integration) and completed its randomized trials. Finally, participants completed Task 3 (subjective evaluation).

5.6 Measures

[TODO: Measures]

5.7 Statistical Analysis

[TODO: Statistical analysis]

6 Results

6.1 Task Performance

[TODO: Task-performance results]

6.2 Subjective Evaluation

[TODO: Subjective-evaluation results]

6.3 Qualitative Feedback

[TODO: Qualitative-feedback results]

7 Discussion

Following Lam et al.'s evaluation framework [20], our study combines understanding of environments and work practices (UWP) through formative expert interviews with controlled evaluation of user performance (UP) and user experience (UE) in the subsequent quantitative study.

Our study design was informed by several pitfalls identified in design study methodology [27]. We aimed to avoid *premature advance* and *premature design commitment* (PF-1, PF-20) by conducting formative interviews with EEG/ERP researchers before finalizing the visualization designs and quantitative tasks, grounding the study in domain practices. We also retained the conventional topoplot as a baseline and preserved familiar aspects of its encoding in several conditions, reflecting the recommendation not to ignore existing practices that already work well (PF-15). Finally, we did not treat subjective preference as sufficient evidence of effectiveness: objective task performance was combined with subjective ratings and open-ended feedback, addressing the concern captured by PF-26. At the same time, the controlled quantitative tasks do not reproduce the full complexity of real EEG/ERP analysis workflows, so the study should not be interpreted as a full real-world case study, consistent with the caution expressed in PF-25.

[TODO: Discussion]

8 Limitations

[TODO: Limitations]

9 Conclusion

[TODO: Conclusion]

References

[1] E. A. Allen, E. B. Erhardt, and V. D. Calhoun, "Data Visualization in the Neurosciences: Overcoming the Curse of Dimensionality," *Neuron*, vol. 74, no. 4, pp. 603–608, 2012, doi: 10.1016/j.neuron.2012.05.001.

[2] J. Bernard, M. Steiger, S. Mittelstädt, S. Thum, D. A. Keim, and J. Kohlhammer, "A Survey and Task-Based Quality Assessment of Static 2D Colormaps," in *Visualization and Data Analysis 2015*, 2015, vol. 9397, p. 93970M, doi: 10.1117/12.2079841.

[3] N. Boukhelifa and D. J. Duke, "Uncertainty Visualization: Why Might It Fail?," in *CHI '09 Extended Abstracts on Human Factors in Computing Systems*, 2009, pp. 4051–4056, doi: 10.1145/1520340.1520616.

[4] K. Brodlie, R. Allendes Osorio, and A. Lopes, "A Review of Uncertainty in Data Visualization," in *Expanding the Frontiers of Visual Analytics and Visualization*, J. Dill, R. Earnshaw, D. Kasik, J. Vince, and P. C. Wong, Eds. London: Springer, 2012, pp. 81–109.

[5] M. Correll, D. Moritz, and J. Heer, "Value-Suppressing Uncertainty Palettes," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 2018, pp. 1–11, doi: 10.1145/3173574.3174216.

[6] S. Danisch and J. Krumbiegel, "Makie.jl: Flexible High-Performance Data Visualization for Julia," *Journal of Open Source Software*, vol. 6, no. 65, p. 3349, 2021, doi: 10.21105/joss.03349.

[7] B. V. Ehinger, "EEGVIS Toolbox," 2018. https://github.com/behinger/eegvis.

[8] S. A. Engel and P. C. Burton, "Confidence Intervals for fMRI Activation Maps," *PLoS ONE*, vol. 8, no. 12, p. e82419, 2013, doi: 10.1371/journal.pone.0082419.

[9] Z. Geršlová, "diegr: Dynamic and Interactive EEG Graphics," 2026. https://gerslovaz.github.io/diegr/reference/plot_topo_mean.html.

[10] Gesellschaft für wissenschaftliche Datenverarbeitung mbH Göttingen (GWDG), "Voice AI." https://docs.hpc.gwdg.de/services/ai-services/voice-ai/.

[11] C. Gillmann, D. Saur, T. Wischgoll, and G. Scheuermann, "Uncertainty-aware Visualization in Medical Imaging: A Survey," *Computer Graphics Forum*, vol. 40, no. 3, pp. 665–689, 2021, doi: 10.1111/cgf.14333.

[12] J. Hullman, "Why Authors Don't Visualize Uncertainty," *IEEE Transactions on Visualization and Computer Graphics*, vol. 26, no. 1, pp. 130–139, 2020, doi: 10.1109/TVCG.2019.2934287.

[13] J. Hullman, P. Resnick, and E. Adar, "Hypothetical Outcome Plots Outperform Error Bars and Violin Plots for Inferences about Reliability of Variable Ordering," *PLOS ONE*, vol. 10, no. 11, p. e0142444, 2015, doi: 10.1371/journal.pone.0142444.

[14] A. Jain, E. Walsh, and A. Lal, "Communicating Uncertainty to Policy Makers in Maps: A Systematic Scoping Review of Meaningful to Misleading Current Practice," *PLOS Climate*, vol. 5, no. 7, p. e0000707, 2026, doi: 10.1371/journal.pclm.0000707.

[15] E. S. Kappenman, J. L. Farrens, W. Zhang, A. X. Stewart, and S. J. Luck, "ERP CORE: An Open Resource for Human Event-Related Potential Research," *NeuroImage*, vol. 225, p. 117465, 2021, doi: 10.1016/j.neuroimage.2020.117465.

[16] M. Kay, "ggdist: Visualizations of Distributions and Uncertainty in the Grammar of Graphics," *IEEE Transactions on Visualization and Computer Graphics*, vol. 30, no. 1, pp. 414–424, 2024, doi: 10.1109/TVCG.2023.3327195.

[17] C. Kinkeldey, A. M. MacEachren, and J. Schiewe, "How to Assess Visual Communication of Uncertainty? A Systematic Review of Geospatial Uncertainty Visualisation User Studies," *The Cartographic Journal*, vol. 51, no. 4, pp. 372–386, 2014, doi: 10.1179/1743277414Y.0000000099.

[18] H. Koo, Y. Chun, and D. A. Griffith, "Geovisualizing Attribute Uncertainty of Interval and Ratio Variables: A Framework and an Implementation for Vector Data," *Journal of Visual Languages and Computing*, vol. 44, pp. 89–96, 2018, doi: 10.1016/j.jvlc.2017.11.007.

[19] P. Kubíček and Č. Šašinka, "Thematic Uncertainty Visualization Usability: Comparison of Basic Methods," *Annals of GIS*, vol. 17, no. 4, pp. 253–263, 2011, doi: 10.1080/19475683.2011.625978.

[20] H. Lam, E. Bertini, P. Isenberg, C. Plaisant, and S. Carpendale, "Empirical Studies in Information Visualization: Seven Scenarios," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 9, pp. 1520–1536, 2012, doi: 10.1109/TVCG.2011.279.

[21] A. M. MacEachren, "Visualizing Uncertain Information," *Cartographic Perspectives*, no. 13, pp. 10–19, 1992, doi: 10.14714/CP13.1000.

[22] V. Mikheev, R. Skukies, and B. V. Ehinger, "The Art of Brainwaves: A Survey on Event-Related Potential Visualization Practices," *Aperture Neuro*, vol. 4, 2024, doi: 10.52294/001c.116386.

[23] L. Padilla, P. S. Quinan, M. Meyer, and S. H. Creem-Regehr, "Evaluating the Impact of Binning 2D Scalar Fields," *IEEE Transactions on Visualization and Computer Graphics*, vol. 23, no. 1, pp. 431–440, 2017, doi: 10.1109/TVCG.2016.2599106.

[24] C. R. Pernet, N. Chauveau, C. Gaspar, and G. A. Rousselet, "LIMO EEG: A Toolbox for Hierarchical LInear MOdeling of ElectroEncephaloGraphic Data," *Computational Intelligence and Neuroscience*, vol. 2011, p. 831409, 2011, doi: 10.1155/2011/831409.

[25] B. Petek, D. Nabergoj, and E. Štrumbelj, "A General Approach to Visualizing Uncertainty in Statistical Graphics," *IEEE Transactions on Visualization and Computer Graphics*, vol. 32, no. 9, pp. 7819–7833, 2026, doi: 10.1109/TVCG.2026.3704459.

[26] J. Sanyal, S. Zhang, G. Bhattacharya, P. Amburn, and R. J. Moorhead, "A User Study to Compare Four Uncertainty Visualization Methods for 1D and 2D Datasets," *IEEE Transactions on Visualization and Computer Graphics*, vol. 15, no. 6, pp. 1209–1218, 2009, doi: 10.1109/TVCG.2009.114.

[27] M. Sedlmair, M. Meyer, and T. Munzner, "Design Study Methodology: Reflections from the Trenches and the Stacks," *IEEE Transactions on Visualization and Computer Graphics*, vol. 18, no. 12, pp. 2431–2440, 2012, doi: 10.1109/TVCG.2012.213.

[28] A. Sterzik, M. Krone, D. Baum, D. W. Cunningham, and K. Lawonn, "Uncertainty Visualization for Biomolecular Structures: An Empirical Evaluation," *IEEE Transactions on Visualization and Computer Graphics*, vol. 31, no. 12, pp. 10296–10310, 2025, doi: 10.1109/TVCG.2025.3596385.

[29] G. Strode *et al.*, "Operationalizing Trumbo's Principles of Bivariate Choropleth Map Design," *Cartographic Perspectives*, no. 94, pp. 5–24, 2020, doi: 10.14714/CP94.1538.

[30] A. J. Teuling, R. Stöckli, and S. I. Seneviratne, "Bivariate Colour Maps for Visualizing Climate Data," *International Journal of Climatology*, vol. 31, no. 9, pp. 1408–1412, 2011, doi: 10.1002/joc.2153.

[31] H. Wainer and C. M. Francolini, "An Empirical Inquiry Concerning Human Understanding of Two-Variable Color Maps," *The American Statistician*, vol. 34, no. 2, pp. 81–93, 1980, doi: 10.1080/00031305.1980.10483006.

[32] C. O. Wilke, *Fundamentals of Data Visualization*. O'Reilly Media, 2019.

[33] R. D. Yanai, T. A. Mann, S. D. Hong, G. Pu, and J. M. Zukswert, "The Current State of Uncertainty Reporting in Ecosystem Studies: A Systematic Evaluation of Peer-Reviewed Literature," *Ecosphere*, vol. 12, no. 6, p. e03535, 2021, doi: 10.1002/ecs2.3535.